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$$z^3 = 8i$$

$$8i = 8\,\mathrm{cis}\,\frac{\pi}{2}$$

$$z_k = \sqrt[3]{8}\,\mathrm{cis}\left(\frac{\frac{\pi}{2} + k2\pi}{3}\right)$$

$$= 2\,\mathrm{cis}\left(\frac{\pi}{6} + k\frac{2\pi}{3}\right)$$

$$z_0 = 2\,\mathrm{cis}\,\frac{\pi}{6} = \sqrt{3} + i$$

$$z_1 = 2\,\mathrm{cis}\,\frac{5\pi}{6} = -\sqrt{3} + i$$

$$z_2 = 2\,\mathrm{cis}\,\frac{3\pi}{2} = -2i$$

References